

Thrust-Vector Control of a Three-Axis Stabilized Spacecraft with Fuel Slosh Dynamics

Jaime Rubio Hervás¹, Mahmut Reyhanoglu^{2*}, and Hui Tang³

^{1,2}Physical Sciences Department
Embry-Riddle Aeronautical University
Daytona Beach, FL 32114, USA

⁴School of Mechanical and Aerospace Engineering
Nanyang Technological University
Singapore 639798

¹rubioj1@my.erau.edu ²reyhanom@erau.edu ³htang@ntu.edu.sg * Corresponding author

Abstract: The thrust vector control problem is studied for an upper-stage rocket with fuel slosh dynamics. The sloshing propellant is modeled as a multi-mass-spring system. The coupled equations of motion of the spacecraft and the fuel are expressed in terms of the three dimensional spacecraft translational velocity vector, the attitude, the angular velocity, and the internal coordinates representing the slosh modes. A Lyapunov-based nonlinear feedback control law is proposed to control the translational velocity vector and the attitude of the spacecraft, while attenuating the sloshing modes. A simulation example is included to illustrate the effectiveness of the control law.

Keywords: Slosh dynamics, Lyapunov methods.

1. INTRODUCTION

The interaction of the fuel slosh dynamics with the rigid body dynamics of the spacecraft is known to have destabilizing effects. A variety of passive techniques have been employed to reduce these adverse effects [1]. The passive techniques do not completely succeed in cancelling the sloshing effects and thus active methods are developed to control the sloshing.

In our previous work [3, 6, 7, 9–12], we studied the nonlinear modeling and control problems for planar maneuvering of space vehicles with fuel slosh dynamics. Both multi-mass-spring models [10, 11], and multi-pendulum models [12] were considered for the characterization of the most prominent sloshing modes. This paper extends our previous work to the three-dimensional maneuvers of a spacecraft with fuel slosh dynamics.

The coupled dynamics of a three-axis stabilized spacecraft with a single fuel tank are formulated by modeling the sloshing propellant as a multi-mass-spring system. Two gimbal deflection angles of a main engine and three independent torques, generated by either gas jet pairs or control moment gyros are assumed to be available as control inputs. A feedback controller is designed to stabilize the translational velocity vector and the attitude dynamics of the spacecraft, while suppressing the sloshing modes.

2. MODEL FORMULATION

This section formulates the dynamics of a three-axis stabilized spacecraft with a single propellant tank including the prominent fuel slosh modes. The spacecraft is represented as a rigid body (base body) and the sloshing fuel masses as internal bodies.

Following the development in [2], let $\mathbf{v} \in \mathcal{R}^3$, $\boldsymbol{\omega} \in \mathcal{R}^3$, and $\mathbf{s} \in \mathcal{R}^N$ denote the base body translational velocity vector, the base body angular velocity vector, and the vector of internal coordinates, respectively. In these variables, the Lagrangian has the form $L = L(\mathbf{v}, \boldsymbol{\omega}, \mathbf{s}, \dot{\mathbf{s}})$, which is $SE(3)$ -invariant in the sense that it does not depend on the base body position and attitude. It is assumed that the internal dissipative forces are derivable from a Rayleigh dissipation function R . Then, the equations of motion of the spacecraft with internal dynamics are shown to be given by [2]:

$$\frac{d}{dt} \frac{\partial L}{\partial \mathbf{v}} + \hat{\boldsymbol{\omega}} \frac{\partial L}{\partial \mathbf{v}} = \boldsymbol{\tau}_t, \quad (1)$$

$$\frac{d}{dt} \frac{\partial L}{\partial \boldsymbol{\omega}} + \hat{\boldsymbol{\omega}} \frac{\partial L}{\partial \boldsymbol{\omega}} + \hat{\mathbf{v}} \frac{\partial L}{\partial \mathbf{v}} = \boldsymbol{\tau}_r, \quad (2)$$

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{\mathbf{s}}} - \frac{\partial L}{\partial \mathbf{s}} + \frac{\partial R}{\partial \dot{\mathbf{s}}} = 0, \quad (3)$$

where $\boldsymbol{\tau}_t \in \mathcal{R}^3$ and $\boldsymbol{\tau}_r \in \mathcal{R}^3$ denote the vectors of generalized control forces (typically from thrusters) and generalized control torques (typically from control moment gyros, reaction wheels, and thrusters), respectively, that act on the base body. In this paper, for a given vector $\mathbf{p} = [p_1, p_2, p_3]^T \in \mathcal{R}^3$, the skew-symmetric matrix $\hat{\mathbf{p}}$ defines the corresponding cross-product operation $\mathbf{p} \times$.

We now derive a multi-mass-spring model of the sloshing fuel where the oscillation frequencies of the mass-spring elements represent the prominent sloshing modes (see e.g., [6, 7, 10, 11]). Consider a rigid spacecraft moving on a three-dimensional space as shown in Fig. 1, where F denotes the constant thrust produced by a gimballed thrust engine. The gimbal deflection angles δ_1 and δ_2 about the spacecraft fixed y -axis and z -axis,

respectively, are considered as control inputs. The input torque $\mathbf{M} = [M_x, M_y, M_z]^T$, generated by gas jet pairs or control moment gyroscopes, is also available for control purposes. For simplicity, xyz axes are assumed to be principal axes. The fluid is modeled by moment of inertia I_0 assigned to a rigidly attached isoinertial mass m_0 and point masses m_i , $i = 1, \dots, N$, whose relative positions along the spacecraft fixed y and z -axis are denoted by ξ_i and η_i , respectively. Moments of inertia of these masses are taken as zero. The locations h_0 and h_i are referenced to the center of mass of undisturbed propellant. Restoring forces $-k_{y_i}\xi_i$ and $-k_{z_i}\eta_i$ act on the mass m_i whenever the mass is displaced from its neutral position $\xi_i = 0$ and $\eta_i = 0$, respectively. Damping constants c_{y_i} and c_{z_i} characterize the dissipative effects of the fuel slosh. In this paper, for simplicity it is assumed that $k_{y_i} = k_{z_i} = k_i$ and $c_{y_i} = c_{z_i} = c_i$. The spacecraft mass m , moments of inertia (J_x, J_y, J_z) , the fuel masses (m_0, m_i) , the distance b between the origin of body xyz axes and the spacecraft center of mass location along the longitudinal axis, and the distance d from the gimbal pivot to the spacecraft center of mass are assumed to be constant. If the tank center is in front of the spacecraft center of mass then $b > 0$. The parameters m_i , h_i , k_i , c_i and b depend on the shape of the fuel tank, the characteristics of the fuel and the fill ratio of the fuel tank. To preserve the static properties of the liquid, the sum of all the masses must be the same as the fuel mass m_f , and the center of mass of the model must be at the same elevation as that of the fuel, i.e.,

$$m_0 + \sum_{i=1}^N m_i = m_f, \quad m_0 h_0 + \sum_{i=1}^N m_i h_i = 0. \quad (4)$$

Denote by $\mathbf{r} = [x, y, z]^T$ the inertial position of the

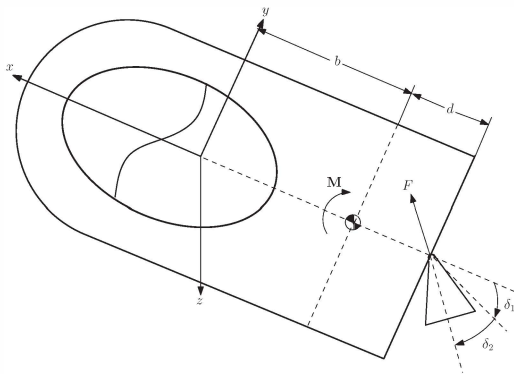


Fig. 1 A spacecraft with a liquid propellant tank.

center of mass of the undisturbed fuel in the spacecraft-fixed coordinate frame. The position vectors of the center of mass of the vehicle and the fuel masses m_0 , m_i , $\forall i$, in the spacecraft-fixed coordinate frame are then given,

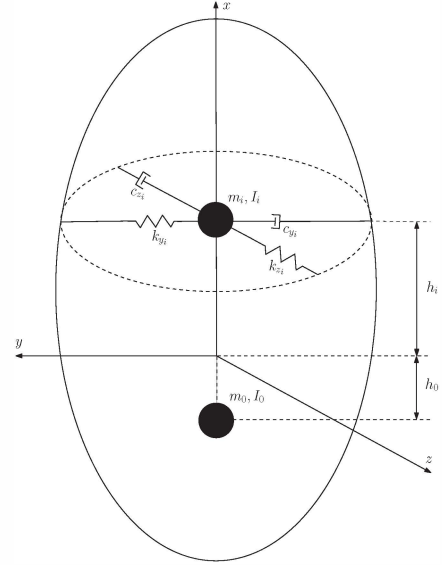


Fig. 2 A multi mass-spring model of sloshing.

respectively, by

$$\mathbf{r}_c = [(x - b), y, z]^T, \quad \mathbf{r}_0 = [x + h_0, y, z]^T, \\ \mathbf{r}_i = [x + h_i, y + \xi_i, z + \eta_i]^T, \quad \forall i.$$

Clearly, the inertial velocity of the center of mass of the undisturbed fuel can be expressed as $\mathbf{v} = \dot{\mathbf{r}} + \boldsymbol{\omega} \times \mathbf{r}$, where $\boldsymbol{\omega} = [\omega_1, \omega_2, \omega_3]^T$ is the angular velocity vector. The inertial velocities $\mathbf{v}_c$, $\mathbf{v}_0$, and $\mathbf{v}_i$, $\forall i$, can be computed similarly.

The total kinetic energy neglecting the moment of inertia of the sloshing masses, can now be expressed as

$$T = \frac{1}{2} m \|\mathbf{v}_c\|^2 + \frac{1}{2} \sum_{j=0}^N m_j \|\mathbf{v}_j\|^2 + \frac{1}{2} \boldsymbol{\omega}^T (\mathbf{J} + I_0 \mathbf{I}) \boldsymbol{\omega},$$

where $\mathbf{J} = \text{diag}\{J_x, J_y, J_z\}$ and $\mathbf{I}$ is the 3×3 identity matrix.

Since gravitational effects are ignored, there is no gravitational potential energy. The acceleration of the rocket gives rise to the elastic potential energy given by

$$U = \frac{1}{2} \sum_{i=1}^N k_i (\xi_i^2 + \eta_i^2).$$

Typically, k_i would be a function of the axial acceleration. However, in this paper it is assumed that the axial acceleration is not significantly affected by small gimbal deflections, attitude changes and fuel motion so that k_i remains constant.

Dissipative effects due to fuel slosh are included via damping constants c_i so that the Rayleigh dissipation function R can be expressed as

$$R = \frac{1}{2} \sum_{i=1}^N c_i (\dot{\xi}_i^2 + \dot{\eta}_i^2).$$

Applying equations (1)-(3) with $L = T - U$ and

$$\mathbf{s} = \begin{bmatrix} \boldsymbol{\xi} \\ \boldsymbol{\eta} \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix}, \quad \boldsymbol{\tau}_t = \begin{bmatrix} F \cos \delta_1 \cos \delta_2 \\ F \sin \delta_2 \\ -F \sin \delta_1 \cos \delta_2 \end{bmatrix},$$

$$\boldsymbol{\omega} = \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix}, \quad \boldsymbol{\tau}_r = \begin{bmatrix} M_x \\ M_y - Fp \sin \delta_1 \cos \delta_2 \\ M_z - Fp \sin \delta_2 \end{bmatrix},$$

where $p = b + d$, the equations of motion can be obtained as

$$(m + m_f)a_x + mb(\omega_2^2 + \omega_3^2) + \sum_{i=1}^N m_i[2\omega_2\dot{\eta}_i - 2\omega_3\dot{\xi}_i + \eta_i(\dot{\omega}_2 + \omega_1\omega_3) - \xi_i(\dot{\omega}_3 - \omega_1\omega_2)] = F \cos \delta_1 \cos \delta_2, \quad (5)$$

$$(m + m_f)a_y - mb(\dot{\omega}_3 + \omega_1\omega_2) + \sum_{i=1}^N m_i[\ddot{\xi}_i - 2\omega_1\dot{\eta}_i - \eta_i(\dot{\omega}_1 - \omega_2\omega_3) - \xi_i(\omega_1^2 + \omega_3^2)] = F \sin \delta_2, \quad (6)$$

$$(m + m_f)a_z + mb(\dot{\omega}_2 - \omega_1\omega_3) + \sum_{i=1}^N m_i[\ddot{\eta}_i + 2\omega_1\dot{\xi}_i - \eta_i(\omega_1^2 + \omega_2^2) - \xi_i(\dot{\omega}_1 + \omega_2\omega_3)] = -F \sin \delta_1 \cos \delta_2, \quad (7)$$

$$(I_0 + J_x)\dot{\omega}_1 + (J_z - J_y)\omega_2\omega_3 + \sum_{i=1}^N m_i[\xi_i\ddot{\eta}_i - \eta_i\ddot{\xi}_i + 2(\eta_i\dot{\eta}_i + \xi_i\dot{\xi}_i)\omega_1 + \eta_i(-a_y + \eta_i(\dot{\omega}_1 - \omega_2\omega_3) - h_i(\dot{\omega}_3 + \omega_1\omega_2)) + \xi_i(a_z + \xi_i(\dot{\omega}_1 + \omega_2\omega_3) - h_i(\dot{\omega}_2 + \omega_1\omega_3))] = M_x, \quad (8)$$

$$(I_0 + J_y)\dot{\omega}_2 + (J_x - J_z)\omega_1\omega_3 + mba_x + \hat{I}(\dot{\omega}_2 - \omega_1\omega_3) + \sum_{i=1}^N m_i[-h_i\ddot{\eta}_i + 2\eta_i\dot{\eta}_i\omega_2 - 2\dot{\xi}_i(\omega_3\eta_i + h_i\omega_1) + \eta_i(a_x + \eta_i(\dot{\omega}_2 + \omega_1\omega_3) + h_i(\omega_1^2 - \omega_3^2)) - h_i\xi_i(\dot{\omega}_1 + \omega_2\omega_3) - \eta_i\xi_i(\dot{\omega}_3 - \omega_1\omega_2)] = M_y - Fp \sin \delta_1 \cos \delta_2, \quad (9)$$

$$(I_0 + J_z)\dot{\omega}_3 + (J_y - J_x)\omega_1\omega_2 - mba_y + \hat{I}(\dot{\omega}_3 + \omega_1\omega_2) + \sum_{i=1}^N m_i[h_i\ddot{\xi}_i + 2\xi_i\dot{\xi}_i\omega_3 - 2\dot{\eta}_i(\omega_2\xi_i + h_i\omega_1) + \xi_i(-a_x + \xi_i(\dot{\omega}_3 - \omega_1\omega_2) + h_i(\omega_2^2 - \omega_1^2)) - h_i\eta_i(\dot{\omega}_1 - \omega_2\omega_3) - \eta_i\xi_i(\dot{\omega}_2 + \omega_1\omega_3)] = M_z - Fp \sin \delta_2, \quad (10)$$

$$m_i[\ddot{\xi}_i + a_y - 2\dot{\eta}_i\omega_1 - \eta_i(\dot{\omega}_1 - \omega_2\omega_3) - \xi_i(\omega_1^2 + \omega_3^2) - h_i(\dot{\omega}_3 - \omega_1\omega_2)] + k_i\xi_i + c_i\dot{\xi}_i = 0, \quad \forall i, \quad (11)$$

$$m_i[\ddot{\eta}_i + a_z + 2\dot{\xi}_i\omega_1 + \xi_i(\dot{\omega}_1 + \omega_2\omega_3) - \eta_i(\omega_1^2 + \omega_2^2) - h_i(\dot{\omega}_2 - \omega_1\omega_3)] + k_i\eta_i + c_i\dot{\eta}_i = 0, \quad \forall i, \quad (12)$$

where $\mathbf{a} = [a_x, a_y, a_z]^T = \dot{\mathbf{v}} + \boldsymbol{\omega} \times \mathbf{v}$ is the acceleration of the center of mass of the fuel in the spacecraft-fixed frame and

$$\hat{I} = mb^2 + m_0h_0^2 + \sum_{i=1}^N m_i h_i^2.$$

Let $\boldsymbol{\theta} = [\theta_1, \theta_2, \theta_3]^T$ denote the vector of Euler angles. Assuming 321 Euler angle sequence, the angular velocity can be expressed as

$$\boldsymbol{\omega} = \mathbf{T}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}}, \quad (13)$$

where

$$\mathbf{T}(\boldsymbol{\theta}) = \begin{bmatrix} 1 & 0 & -\sin \theta_2 \\ 0 & \cos \theta_1 & \sin \theta_1 \cos \theta_2 \\ 0 & -\sin \theta_1 & \cos \theta_1 \cos \theta_2 \end{bmatrix}.$$

The control objective is to design feedback controllers so that the controlled spacecraft accomplishes a given three-dimensional maneuver while suppressing the fuel slosh modes.

3. FEEDBACK CONTROLLER DESIGN

Consider the model of a spacecraft with a gimbaled thrust engine shown in Fig. 1-2. If the thrust F during the fuel burn is a positive constant, and if the gimbal deflection angles and moments are zero, $\delta_1 = \delta_2 = M_x = M_y = M_z = 0$, then the spacecraft and fuel slosh dynamics have a relative equilibrium defined by

$$v_y = \bar{v}_y, \quad v_z = \bar{v}_z, \quad \boldsymbol{\theta} = \bar{\boldsymbol{\theta}}, \quad \dot{\boldsymbol{\theta}} = 0, \quad \mathbf{s} = \dot{\mathbf{s}} = 0,$$

where $\bar{v}_y, \bar{v}_z$ and $\bar{\boldsymbol{\theta}}$ are arbitrary constants. Without loss of generality, the subsequent analysis considers the relative equilibrium at the origin, i.e., $\bar{v}_y = \bar{v}_z = \bar{\boldsymbol{\theta}} = 0$. Note that the relative equilibrium corresponds to the vehicle axial velocity

$$v_x(t) = v_{x_0} + \bar{a}_x t, \quad t \leq t_b,$$

where v_{x_0} is the initial axial velocity of the spacecraft, t_b is the fuel burn time, and $\bar{a}_x = F/(m + m_f)$.

Now assume the axial acceleration term a_x is not significantly affected by small gimbal deflections, attitude changes and fuel motion (an assumption verified in simulations). Consequently, equation (5) becomes:

$$\dot{v}_x + \omega_2 v_z - \omega_3 v_y = \bar{a}_x. \quad (14)$$

Substituting this approximation into (6)-(12) leads to the following reduced equations of motion for the transverse, attitude and slosh dynamics where $v_x(t)$ is considered as an exogenous input. The subsequent analysis is based on these equations of motion for the transverse, attitude and slosh dynamics of the vehicle.

We now design a nonlinear controller to stabilize the relative equilibrium at the origin of the equations (6)-(12). Our control design is based on a Lyapunov function approach. By defining control transformations from $(\delta_1, \delta_2, M_x, M_y, M_z)$ to new control inputs $(u_1, u_2, u_3, u_4, u_5)$, the equations (6)-(12) can be writ-

ten as:

$$\dot{v}_y = u_1 - v_x \omega_3 + v_z \omega_1, \quad (15)$$

$$\dot{v}_z = u_2 + v_x \omega_2 - v_y \omega_1, \quad (16)$$

$$\ddot{\theta}_1 = u_3, \quad (17)$$

$$\ddot{\theta}_2 = u_4, \quad (18)$$

$$\ddot{\theta}_3 = u_5, \quad (19)$$

$$\begin{aligned} \dot{\xi}_i = & -\Omega_i^2 \xi_i - 2\zeta_i \Omega_i \dot{\xi}_i - u_1 + 2\dot{\eta}_i \omega_1 \\ & + \eta_i (\dot{\omega}_1 - \omega_2 \omega_3) + \xi_i (\omega_1^2 + \omega_3^2) + h_i (\dot{\omega}_3 - \omega_1 \omega_2), \end{aligned} \quad (20)$$

$$\begin{aligned} \dot{\eta}_i = & -\Omega_i^2 \eta_i - 2\zeta_i \Omega_i \dot{\eta}_i - u_2 - 2\dot{\xi}_i \omega_1 \\ & - \xi_i (\dot{\omega}_1 + \omega_2 \omega_3) + \eta_i (\omega_1^2 + \omega_2^2) + h_i (\dot{\omega}_2 - \omega_1 \omega_3), \end{aligned} \quad (21)$$

where

$$\Omega_i^2 = \frac{k_i}{m_i}, \quad 2\zeta_i \Omega_i = \frac{c_i}{m_i}, \quad \forall i.$$

Here Ω_i and ζ_i , $\forall i$, denote the undamped natural frequencies and damping ratios, respectively.

Now, consider the following candidate Lyapunov function for the system (15)-(21):

$$\begin{aligned} V = & \frac{r_1}{2} v_y^2 + \frac{r_2}{2} v_z^2 + \frac{r_3}{2} \theta_1^2 + \frac{r_4}{2} \dot{\theta}_1^2 + \frac{r_5}{2} \theta_2^2 + \frac{r_6}{2} \dot{\theta}_2^2 + \frac{r_7}{2} \theta_3^2 \\ & + \frac{r_8}{2} \dot{\theta}_3^2 + \frac{r_9}{2} \sum_{i=1}^N (\dot{\xi}_i^2 + \Omega_i^2 \xi_i^2 - 2h_i \dot{\xi}_i \omega_3 - 2\eta_i \dot{\xi}_i \omega_1) \\ & + \frac{r_{10}}{2} \sum_{i=1}^N (\dot{\eta}_i^2 + \Omega_i^2 \eta_i^2 - 2h_i \dot{\eta}_i \omega_2 + 2\zeta_i \dot{\eta}_i \omega_1), \end{aligned} \quad (22)$$

where r_i , $i = 1, \dots, 10$, are positive constants. Assume that $r_4 = r_6 = r_8 \gg 1 \gg r_9 = r_{10}$. so that the function V is positive definite.

The time derivative of V along the trajectories of (15)-(21) is

$$\begin{aligned} \dot{V} = & -2r_9 \sum_{i=1}^N \zeta_i \Omega_i \dot{\xi}_i^2 - 2r_{10} \sum_{i=1}^N \zeta_i \Omega_i \dot{\eta}_i^2 \\ & + [r_1 v_y - r_9 \sum_{i=1}^N (\dot{\xi}_i - h_i \omega_3 - \eta_i \omega_1)] u_1 \\ & + [r_2 v_z - r_{10} \sum_{i=1}^N (\dot{\eta}_i - h_i \omega_2 + \xi_i \omega_1)] u_2 \\ & + \sum_{i=1}^3 (f_{i1} u_3 + f_{i2} u_4 + f_{i3} u_5 + g_i) \dot{\theta}_i, \end{aligned} \quad (23)$$

where f_{ij} , g_i , $i, j = 1, 2, 3$, are functions of v_y , v_z , θ , $\dot{\theta}$, s , and $\dot{s}$.

Assume $\mathbf{F} = [f_{ij}]$ is an invertible matrix $\forall t$. Then, the

feedback laws

$$u_1 = -l_1 [r_1 v_y - r_9 \sum_{i=1}^N (\dot{\xi}_i - h_i \omega_3 - \eta_i \omega_1)], \quad (24)$$

$$u_2 = -l_2 [r_2 v_z - r_{10} \sum_{i=1}^N (\dot{\eta}_i - h_i \omega_2 + \xi_i \omega_1)], \quad (25)$$

$$\begin{bmatrix} u_3 \\ u_4 \\ u_5 \end{bmatrix} = -\mathbf{F}^{-1} \begin{bmatrix} g_1 + l_3 \dot{\theta}_1 \\ g_2 + l_4 \dot{\theta}_2 \\ g_3 + l_5 \dot{\theta}_3 \end{bmatrix}, \quad (26)$$

where l_i , $i = 1, \dots, 5$, are positive constants, yield

$$\begin{aligned} \dot{V} = & -2r_9 \sum_{i=1}^N \zeta_i \Omega_i \dot{\xi}_i^2 - 2r_{10} \sum_{i=1}^N \zeta_i \Omega_i \dot{\eta}_i^2 \\ & - l_1 [r_1 v_y - r_9 \sum_{i=1}^N (\dot{\xi}_i - h_i \omega_3 - \eta_i \omega_1)]^2 \\ & - l_2 [r_2 v_z - r_{10} \sum_{i=1}^N (\dot{\eta}_i - h_i \omega_2 + \xi_i \omega_1)]^2 \\ & - l_3 \dot{\theta}_1^2 - l_4 \dot{\theta}_2^2 - l_5 \dot{\theta}_3^2, \end{aligned}$$

which satisfies $\dot{V} \leq 0$. Using LaSalle's principle, it is easy to prove asymptotic stability of the origin of the closed loop defined by equations (15)-(21) and the feedback control laws (24)-(26).

Remark: If we assume small Euler angles, the formulation above can be made more tractable. In this case, $\omega = \dot{\theta}$, and f_{ij} , g_i , $i = 1, 2, 3$, take the following form:

$$f_{11} = r_4 - \sum_{i=1}^N (r_9 \eta_i^2 + r_{10} \xi_i^2), \quad f_{12} = f_{21} = r_{10} \sum_{i=1}^N h_i \xi_i,$$

$$f_{22} = r_6 - r_{10} \sum_{i=1}^N h_i^2, \quad f_{13} = f_{31} = -r_9 \sum_{i=1}^N h_i \eta_i,$$

$$f_{33} = r_8 - r_9 \sum_{i=1}^N h_i^2, \quad f_{23} = f_{32} = 0,$$

$$\begin{aligned} g_1 = & (r_1 - r_2) v_y v_z + r_3 \theta_1 + r_9 \sum_{i=1}^N [2\dot{\xi}_i \dot{\eta}_i + \dot{\theta}_1 \xi_i \dot{\xi}_i - \dot{\theta}_2 h_i \dot{\xi}_i \\ & - 2(h_i \dot{\theta}_3 + \dot{\theta}_1 \eta_i) \dot{\eta}_i + \Omega_i^2 \xi_i \eta_i + 2\zeta_i \Omega_i \eta_i \dot{\xi}_i] - r_{10} \sum_{i=1}^N [2\dot{\xi}_i \dot{\eta}_i \\ & - \dot{\theta}_1 \eta_i \dot{\eta}_i + \dot{\theta}_3 h_i \dot{\eta}_i - 2(h_i \dot{\theta}_2 - \dot{\theta}_1 \xi_i) \dot{\xi}_i + \Omega_i^2 \xi_i \eta_i + 2\zeta_i \Omega_i \xi_i \dot{\eta}_i], \end{aligned}$$

$$\begin{aligned} g_2 = & r_2 v_x v_z + r_5 \theta_2 - r_{10} \sum_{i=1}^N (\dot{\theta}_3 \xi_i \dot{\eta}_i - \dot{\theta}_2 \eta_i \dot{\eta}_i - h_i \Omega_i^2 \eta_i \\ & - 2h_i \zeta_i \Omega_i \dot{\eta}_i) - r_9 \sum_{i=1}^N \dot{\theta}_3 \eta_i \dot{\xi}_i, \end{aligned}$$

$$g_3 = -r_1 v_x v_y + r_7 \theta_3 + r_9 \sum_{i=1}^N (\dot{\theta}_3 \xi_i \dot{\xi}_i + h_i \Omega_i^2 \xi_i + 2h_i \zeta_i \Omega_i \xi_i \dot{\xi}_i).$$

4. SIMULATION

The feedback control laws developed in the previous section are implemented here for the AVUM upper stage spacecraft [8]. The first two slosh modes are included to demonstrate the effectiveness of the control law. The physical parameters used in the simulation is given in Table 1. The fluid parameters are obtained using the formulae in [4]. The control objective is to stabilize the relative equilibrium corresponding to a constant axial spacecraft acceleration of 1.720 m/s^2 and $v_y = v_z = 0$, $\theta = \dot{\theta} = 0$, $s = \dot{s} = 0$.

Table 1 Spacecraft and slosh parameters for AVUM.

m	975 kg	F	2450 N
J_x	$600 \text{ kg} \cdot \text{m}^2$	t_b	600 s
J_y	$800 \text{ kg} \cdot \text{m}^2$	b	-0.6 m
J_z	$400 \text{ kg} \cdot \text{m}^2$	d	1.2 m
m_0	358 kg	h_0	-0.011 m
m_1	89 kg	h_1	0.035 m
m_2	2.7 kg	h_2	0.291 m
I_0	$14.85 \text{ kg} \cdot \text{m}^2$	$\bar{a}_x$	1.720 m/s^2
k_1	750 N/m	c_1	$25.8 \text{ N} \cdot \text{s/m}$
k_2	65 N/m	c_2	$1.32 \text{ N} \cdot \text{s/m}$

The effectiveness of the Lyapunov-based control laws is demonstrated by applying the controller defined by (24)-(26) with parameters (24)-(26) with parameters $r_1 = r_2 = 8 \times 10^{-7}$, $r_3 = r_5 = r_7 = 1000$, $r_4 = r_6 = r_8 = 500$, $r_9 = r_{10} = 1 \times 10^{-5}$, $l_1 = l_2 = 1 \times 10^4$, $l_3 = l_4 = l_5 = 4 \times 10^4$ to the complete nonlinear system (5)-(13).

Time responses shown in Figs. 3-8 correspond to the initial conditions

$$\mathbf{v}_0 = [3000, 75, 25]^T \text{ m/s}, \quad \boldsymbol{\theta}_0 = [5^\circ, 2^\circ, -5^\circ]^T,$$

$$\mathbf{s}_0 = [0.15, 0.05, -0.1, 0.15]^T \text{ m}, \quad \dot{\boldsymbol{\theta}}_0 = \mathbf{0}, \quad \dot{\mathbf{s}}_0 = \mathbf{0}.$$

As can be seen in the figures, the transverse velocities, attitude angles, and the slosh states converge to the relative equilibrium at zero while the axial velocity v_x increases and $\dot{v}_x$ tends asymptotically to 1.720 m/s^2 .

5. CONCLUSIONS

A complete nonlinear dynamical model has been developed for three-axis stabilized space vehicles with multiple slosh modes. A Lyapunov-based nonlinear feedback control law has been designed to control the translational velocity vector and the attitude of the spacecraft, while attenuating the sloshing modes characterizing the internal dynamics. A simulation example has been included to illustrate the effectiveness of the control law. Future research includes formulation of problems involving multiple propellant tanks. The control system design in this paper is based on the assumption that the slosh parameters do not significantly change in time. Relaxing this assumption presents a promising avenue of research.

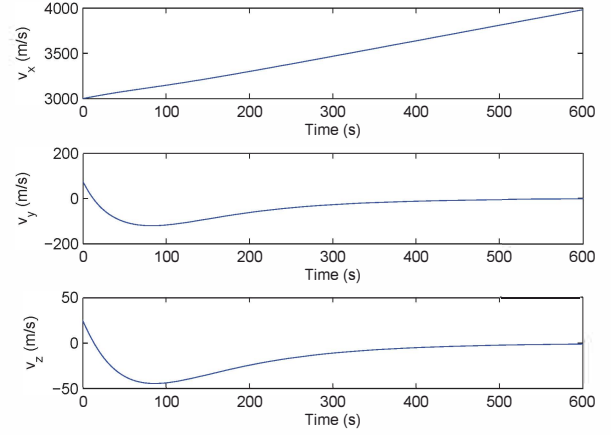


Fig. 3 Time responses of v_x , v_y and v_z .

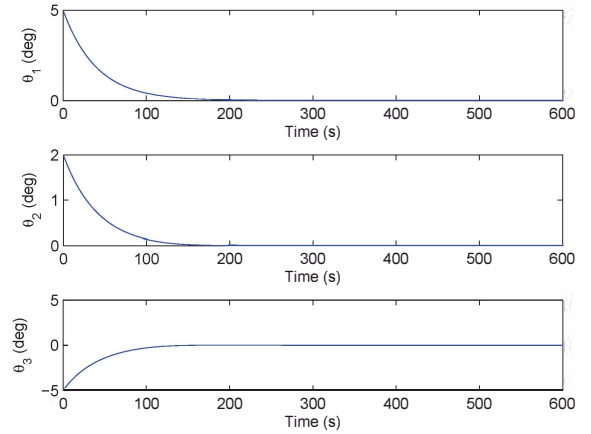


Fig. 4 Time responses of θ_1 , θ_2 and θ_3 .

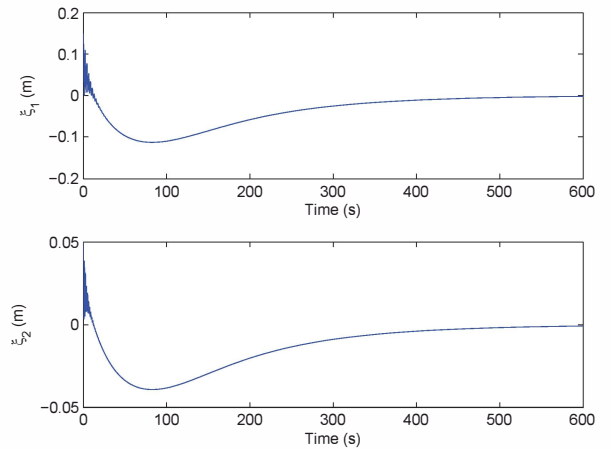


Fig. 5 Time responses of ξ_1 and ξ_2 .

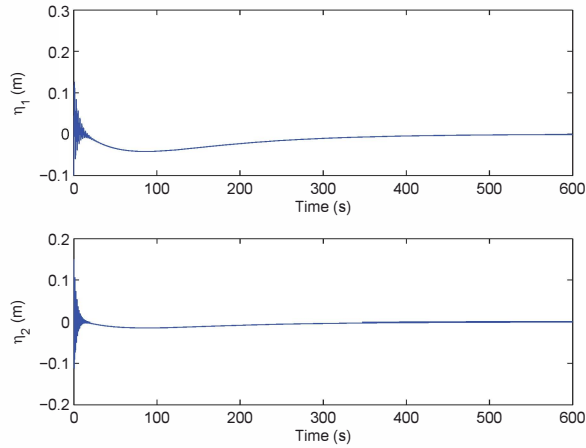


Fig. 6 Time responses of η_1 and η_2 .

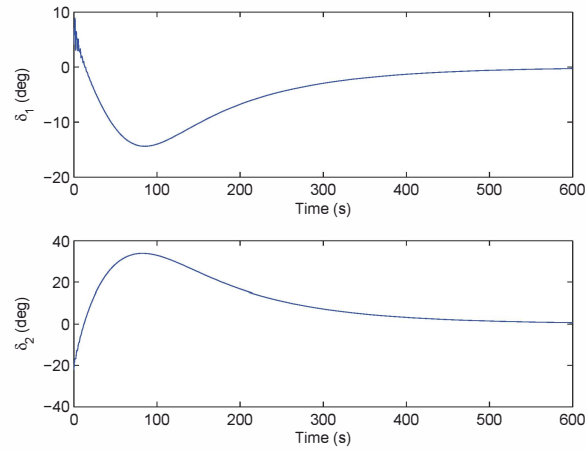


Fig. 7 Gimbal deflection angles δ_1 and δ_2 .

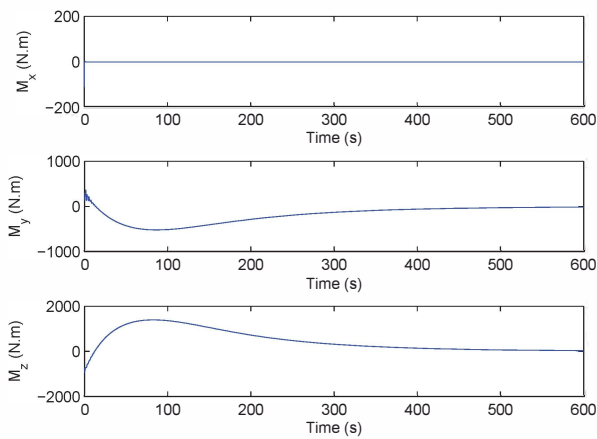


Fig. 8 Control moments M_x , M_y and M_z .

ACKNOWLEDGEMENTS

The authors wish to acknowledge the support provided by Nanyang Technological University.

REFERENCES

- [1] C. Biswal, S. K. Bhattacharyya, and P. K. Sinha, "Dynamic Characteristics of Liquid Filled Rectangular Tank with Baffles," *IE (I) Journal-CV*, Vol. 84, pp. 145-148, 2003.
- [2] S. Cho, N. H. McClamroch, and M. Reyhanoglu, "Dynamics of Multibody Vehicles and Their Formulation as Nonlinear Control Systems," *Proceedings of American Control Conference*, 2000, pp. 3908-3912.
- [3] S. Cho, N. H. McClamroch, and M. Reyhanoglu, "Feedback Control of a Space Vehicle with Unactuated Fuel Slosh Dynamics," *Proceedings of AIAA Guidance, Navigation, and Control Conference*, Vol.1, 2000, AIAA 2000-4046, pp. 354-359.
- [4] F. T. Dodge, "The New Dynamic Behavior of Liquids in Moving Containers," Southwest Research Institute, San Antonio, TX, 2000.
- [5] P.J. Enright and E.C. Wong, "Propellant Slosh Models for the Cassini Spacecraft," *Proceedings of AIAA/AAS Conference*, AIAA-94-3730-CP, 1994, pp. 186-195.
- [6] J. Rubio Hervas and M. Reyhanoglu, "Thrust Vector Control of an Upper-Stage Rocket with Multiple Fuel Slosh Modes," *Mathematical Problems in Engineering*, Vol. 2012, Article ID 848741, 18 pages, 2012.
- [7] J. Rubio Hervas and M. Reyhanoglu, "Control of a Spacecraft with Time-Varying Propellant Slosh Parameters," *Proceedings of International Conference on Control, Automation and Systems*, 2012, pp. 1621-1626.
- [8] E. Perez, *Vega User's Manual*, Issue 3, Arianespace, 2006.
- [9] "Maneuvering Control Problems for a Spacecraft with Unactuated Fuel Slosh Dynamics," *Proceedings of IEEE Conference on Control Applications*, 2003, pp. 695-699.
- [10] M. Reyhanoglu and J. Rubio Hervas, "Nonlinear Dynamics and Control of Space Vehicles with Multiple Fuel Slosh Modes," *IFAC Journal Control Engineering Practice*, Vol. 20, pp. 912-918, 2012.
- [11] M. Reyhanoglu and J. Rubio Hervas, "Nonlinear Control of Space Vehicles with Multi-Mass Fuel Slosh Dynamics," *Proceedings of International Conference on Recent Advances in Space Technologies*, 2011, pp. 247-252.
- [12] M. Reyhanoglu and J. Rubio Hervas, "Nonlinear Control of a Spacecraft with Multiple Fuel Slosh Modes," *Proceedings of IEEE Conference on Decision and Control*, 2011, pp. 6192-6197.